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Original article

Effect of preoperative medial meniscus status on the outcomes of high tibial osteotomy with human umbilical cord-derived mesenchymal stem cells cartilage regeneration

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ABSTRACT

Background: The effect of medial meniscus (MM) status on outcomes following high tibial osteotomy (HTO) combined with cartilage regeneration using human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs) remains unclear.

Purpose: This study aimed to evaluate the effect of preoperative MM status on the outcomes of HTO combined with cartilage regeneration using hUCB-MSCs. Specifically, clinical and radiological outcomes were compared between two groups of patients, which were divided according to their preoperative MM status. We hypothesized that patients with preserved meniscal integrity or those who underwent meniscal root repair would show better clinical and radiological outcomes compared to those with significant meniscal loss or untreated root tears.

Methods: A retrospective analysis was performed on 47 patients who underwent HTO with hUCB-MSC implantation. Patients were divided into group P (preserved MM integrity or medial meniscal root repair) and group L (loss of MM integrity, defined as a peripheral rim width <3 mm, unable to maintain hoop function). Clinical outcomes were assessed using the International Knee Documentation Committee (IKDC) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) scores after a minimum follow-up of 2 years. Cartilage regeneration was evaluated with Magnetic Resonance Observation of Cartilage Repair Tissue (MOCART) 2.0 score and International Cartilage Repair Society (ICRS) Cartilage Repair Assessment (CRA) score from second-look arthroscopy. Correlation analyses were conducted to examine the relationship between preoperative MM extrusion and cartilage repair.

Results: Both groups demonstrated significant improvements in IKDC and WOMAC scores ($p < 0.01$). No significant differences were observed between groups in IKDC and WOMAC scores at final follow-up ($p = 0.21$, $p = 0.42$, respectively). MOCART 2.0 and ICRS CRA scores showed no significant differences between groups ($p = 0.35$, $p = 0.08$, respectively). Group P showed higher proportions of favorable outcomes compared to group L, including no major subchondral changes or only minor marrow edema (56% vs. 31.8%) and ICRS CRA grades I or II (84% vs. 72.7%). While these findings suggest potential differences in outcomes, neither comparison reached statistical significance ($p = 0.09$ and $p = 0.48$, respectively). Preoperative MM extrusion negatively correlated with MOCART 2.0 and subchondral bone changes ($r = -0.24$, $p = 0.03$; $r = -0.29$, $p = 0.02$, respectively).

Conclusion: HTO with hUCB-MSC implantation provided significant clinical improvements and effective cartilage regeneration regardless of preoperative MM status. However, preoperative MM extrusion may influence subchondral bone changes, emphasizing the need to consider MM status for long-term outcomes.

Level of evidence: III; retrospective comparative study

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1. Introduction

High tibial osteotomy (HTO), a well-established surgical procedure, is primarily indicated for patients with varus deformity and medial compartment osteoarthritis (OA) or subchondral insufficiency fractures of the knee [1]. The main objectives of HTO are alleviating knee pain and slowing the progression of degenerative changes by redistributing mechanical load away from the medial compartment [1]. In patients undergoing HTO, it is common to encounter concomitant cartilage damage in the medial compartment, and in patients with extensive cartilage defects, concurrent cartilage regeneration procedures are often performed [1,2]. Cartilage regeneration procedures, including microfracture and autologous chondrocyte implantation (ACI), are frequently employed with HTO to potentially enhance long-term outcomes [3–5]. While microfracture and ACI have demonstrated efficacy in managing focal cartilage lesions, their applicability in OA treatment remains limited [3,6,7].

The use of mesenchymal stem cells (MSCs) with HTO has recently emerged as a promising approach for cartilage restoration in patients with OA [8–11]. MSCs, known for their anti-inflammatory and chondrogenic properties, can be derived from tissues, e.g., bone marrow, synovium, and adipose tissue. However, these traditional sources often require invasive collection methods [12]. Human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs) offer several advantages, including non-invasive harvesting, high expansion potential, and low immunogenicity [12], and as an allogeneic, off-the-shelf product, hUCB-MSCs provide a readily available supply of high-quality stem cells for cartilage repair [8,10,11,13]. hUCB-MSCs are effective and safe for cartilage regeneration, particularly in patients with OA, without causing adverse events [13,14]. This makes the use of hUCB-MSCs with HTO a promising approach for improving long-term outcomes.

Previous studies on HTO combined with cartilage regeneration using hUCB-MSCs have not specifically investigated the effect of medial meniscus (MM) status on treatment outcomes [10,11,15]. The MM plays a key role in shock absorption, joint stability, and the protection of articular cartilage [16]; thus, its absence can substantially compromise the benefits of HTO by accelerating the progression of medial compartment OA [17,18]. In the biomechanics of the knee, proper alignment is essential for optimal extra-articular load distribution, and intra-articularly, the integrity of the meniscus is crucial for maintaining joint function [19,20]. Recent studies have highlighted the importance of preserving meniscal function during HTO to ensure that these protective and stabilizing effects are maintained [21–24].

This study aimed to evaluate the effect of preoperative MM status on the outcomes of HTO combined with cartilage regeneration using hUCB-MSCs. Specifically, clinical and radiological outcomes were compared between two groups of patients, which were divided according to their preoperative MM status. We hypothesized that patients with preserved meniscal integrity or those who underwent meniscal root repair would show better clinical and radiological outcomes compared to those with significant meniscal loss or untreated root tears.

2. Patients and methods

2.1. Study population and design

This retrospective study was conducted at a single institution following approval from the Institutional Review Board and Ethics Committee (IRB No. KUMC 2023-09-056), and the procedures were performed with patients' informed consent. We reviewed 54 patients who underwent HTO with cartilage repair using hUCB-MSCs between March 2017 and December 2021. Specifically, this study screened patients who had plate removal and second-look arthroscopy to evaluate cartilage regeneration after achieving complete bony union, with a minimum follow-up period of 1 year. Inclusion criteria were full-

thickness cartilage defects (International Cartilage Repair Society [ICRS] grade $\geq 3B$) in the medial femoral condyle (MFC), Kellgren–Lawrence (K–L) grade $\geq II$ on Rosenberg view (45° posteroanterior flexion weight-bearing radiograph), and a mechanical axis of hip-knee-ankle angle of $>5^\circ$ varus on a whole-leg weight-bearing anteroposterior radiograph. Exclusion criteria were severe obesity (body mass index $>35 \text{ kg/m}^2$), previous ipsilateral knee fractures, prior cartilage repair procedures, presence of ICRS grade ≥ 3 cartilage lesions in the lateral or patellofemoral compartments, MFC cartilage defects $<2 \text{ cm}^2$, concurrent HTO with cruciate ligament reconstruction or meniscal allograft transplantation, <2 -year follow-up, KL grade IV with definite intra-articular deformities, and congenital deformities leading to malalignment. Out of 54 patients, 47 met the inclusion criteria and were categorized into two groups according to the preoperative condition of the MM. Group P ($n = 25$) comprised patients with a peripheral rim of the medial meniscus measuring 3 mm or more or those who underwent medial meniscal root repair, whereas group L ($n = 21$) included patients with a peripheral rim of the medial meniscus measuring less than 3 mm or those with an untreated medial meniscal root tear (Fig. 1). The criterion of 3 mm was based on existing literature and represents the threshold below which the meniscus cannot maintain its hoop function [25,26].

2.2. Surgical techniques and postoperative rehabilitation

Prior to HTO, hUCB-MSCs (Cartistem®, Medipost Inc., Seoul, Republic of Korea), which were approved by the Korea Food and Drug Administration in January 2012, were prepared. The product includes vials containing 1.5-mL hUCB-MSCs (7.5×10^6 cells) and 4% lyophilized hyaluronic acid (HA) hydrogel. To prepare the mixture, 1.5 mL of the cells were drawn into a sterile syringe and added directly to the freeze-dried HA hydrogel. Then, the mixture was left at room temperature for 30 min before being transferred to a syringe for implantation, following the manufacturer's instructions. Next, the therapeutic dose of hUCB-MSCs was determined on the basis of the cartilage defect size in the MFC assessed by magnetic resonance imaging (MRI). One vial was used for defects $<3 \text{ cm}^2$, whereas two vials were used for defects $\geq 3 \text{ cm}^2$.

All surgical procedures were performed by a single experienced surgeon (D.W.L.), and the arthroscopic implantation of hUCB-MSCs was routinely conducted before HTO. To assess MM integrity, the peripheral rim was measured using a probe, and the hoop tension loss was evaluated based on a threshold of 3 mm in specific regions [25,26]. During the procedure, the calcified cartilage layer was thoroughly removed using an open curette and a shaver, and then multiple drill holes were created at the cartilage defect sites of the MFC using a customized drill bit (Medipost Inc., Seoul, Republic of Korea). The primary drill holes (diameter, 3 mm; depth, 10 mm) were spaced approximately 2 mm apart, with bone ridges left between them. Additionally, smaller holes (diameter, 2 mm; depth, 10 mm) were made in the spaces between the primary holes. Subsequently, the hUCB-MSCs and HA hydrogel composite was carefully implanted into the drill holes, filling the cartilage defects from the base to the surface (Fig. 2).

The surgical procedure for the biplanar opening HTO was conducted using an established technique [27,28]. The correction angle was preoperatively planned to achieve the target alignment through the Fujisawa point, i.e., 62.5% of the tibial plateau width from the medial edge on a full-length weight-bearing anteroposterior radiograph. Then, the osteotomy site was stabilized using a locking plate system (Tomofix, DePuy Synthes, Raynham, MA, USA).

Patients were allowed partial weight-bearing with a web brace (OA Reaction Web Knee Brace, DonJoy OA Adjuster, DJO Global, Vista, CA, USA) for 12 weeks. For the first 3 weeks, toe-touch weight-bearing was allowed, and then weight-bearing was increased to 30% until the 6-week mark, followed by a 30% increase every 3 weeks thereafter. The patients were advised to start range of motion exercises using a continuous

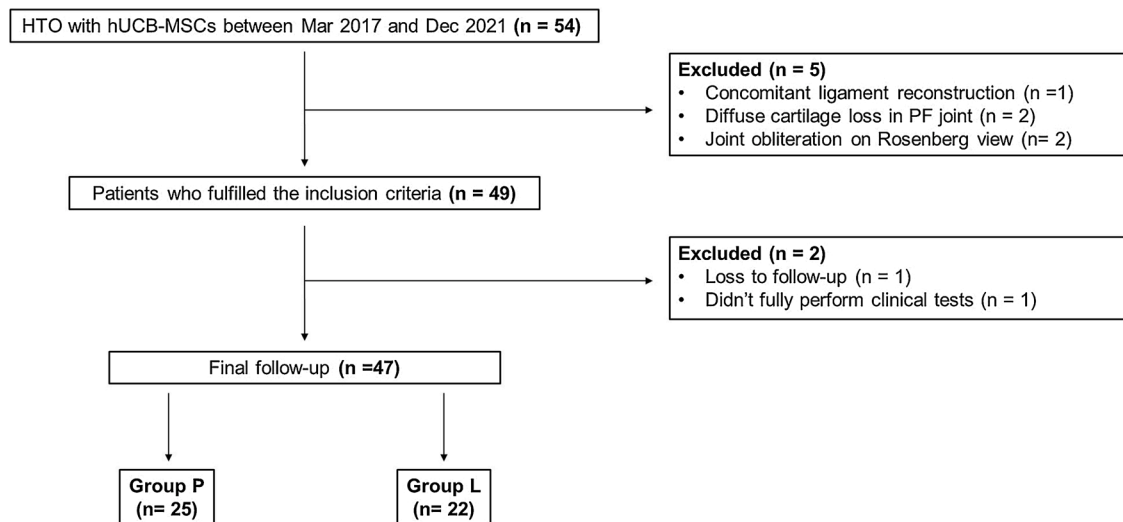


Fig. 1. Flowchart of the included patients. HTO, high tibial osteotomy; hUCB-MSCs, human umbilical cord blood-derived mesenchymal stem cells; PF, patellofemoral; Group P, patients with a peripheral rim of the medial meniscus measuring 3 mm or more or who underwent medial meniscal root repair; Group L, patients with a peripheral rim of the medial meniscus measuring less than 3 mm or who had an untreated medial meniscal root tear.

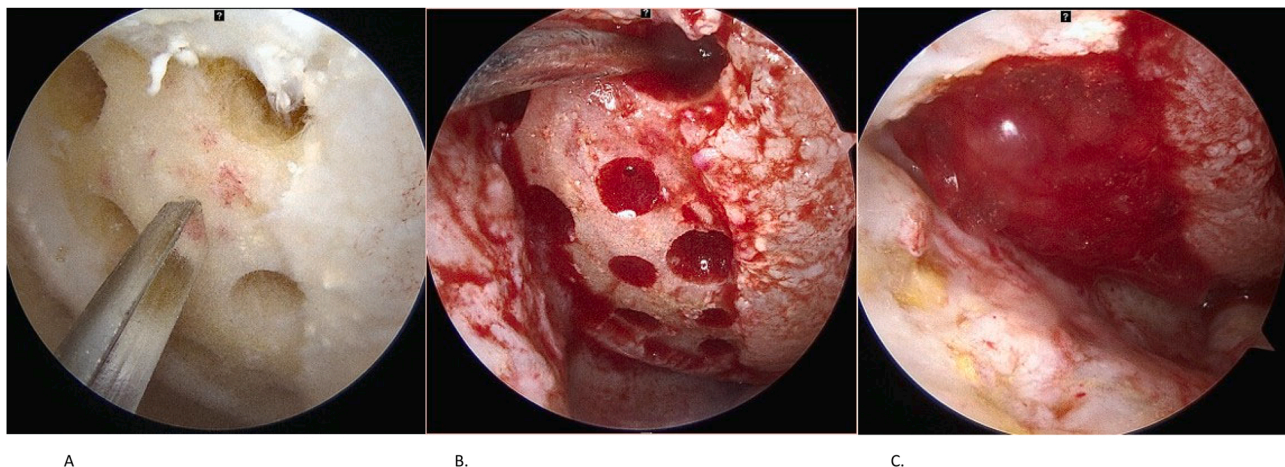


Fig. 2. The process of arthroscopic implantation of human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs). **A:** Multiple drill holes were created at the cartilage defect sites of the medial femoral using a customized drill bit (Medipost Inc., Seoul, Republic of Korea). The drill holes (diameter, 2–3 mm; depth, 10 mm) were spaced approximately 2 mm apart. **B:** The hUCB-MSCs and hyaluronic acid hydrogel composite was implanted into the drill holes from the base to the surface. **C:** The transplanted hUCB-MSCs were confirmed to remain in place even after range of motion.

passive motion (CPM) device on the third day postoperatively, beginning at 30° and increasing by 30° each week until reaching 120°. Additionally, they were instructed to use the CPM device for a total of 6 weeks, with a daily duration of 6 h. Isometric quadriceps exercises were initiated on the first day postoperatively, with squat exercises starting at 3 months. Light running was permitted at 6 months, and return to noncontact sports was allowed at 9 months postoperatively, although strenuous contact sports remained prohibited.

2.3. Clinical evaluations

All patients underwent subjective functional evaluation using the International Knee Documentation Committee (IKDC) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) scores preoperatively and at the final follow-up.

Isokinetic muscle strength was measured using a Biodex System IV dynamometer (Biodex Medical Systems, Shirley, NY) at 60°/s angular velocity, and peak knee extension torque (Nm) was recorded for the involved and uninvolved knees. Each measurement was performed

thrice, and the average value was used in the analysis. Subsequently, tide-to-side torque deficits (%) were calculated to compare muscle strength between the two sides.

2.4. Radiological evaluations

Preoperative and follow-up radiographic assessments were consistently conducted at 3, 6, and 12 months postoperatively, with annual evaluations performed thereafter. The progression of medial compartment OA was determined using the Kellgren–Lawrence (K–L) grading system. Radiographic analysis was based on the Rosenberg view, a 45° posteroanterior flexion weight-bearing radiograph, which is more effective in detecting osteoarthritic progression than the full extension weight-bearing radiograph [29,30]. Lower extremity alignment was assessed by measuring the mechanical axis of the hip–knee–ankle angle using full-length, weight-bearing anteroposterior radiographs; this measurement is crucial for understanding load distribution across the knee joint and the potential progression of joint pathology. Two experienced professionals independently performed radiological

measurements—an orthopedic surgeon and a radiologist (D.W.L and S. G.M., respectively). To enhance measurement reliability, each examiner conducted the measurements on two separate occasions, scheduled 6 weeks apart.

To assess the condition of the medial compartment cartilage and the MM, MRI was conducted preoperatively and at 12 months postoperatively using a 3.0-T system (Signa HD, GE Healthcare, Milwaukee, WI, USA). The postoperative MRI examination was typically performed 1–2 weeks before the second-look arthroscopy. Cartilage regeneration was evaluated using the Magnetic Resonance Observation of Cartilage Repair Tissue (MOCART) 2.0 knee score on MRI, and seven specific variables from the MOCART 2.0 knee score were analyzed to assess the characteristics of the repaired tissue [31]. Specifically, meniscal extrusion was assessed as an indicator of meniscal function, measured on coronal MRI images as the distance from the medial edge of the MM to the superomedial aspect of the tibial plateau. Extrusion exceeding 3 mm was defined as pathological [32,33]. Preoperative MRI was used to assess meniscal extrusion, while postoperative MRI focused on evaluating cartilage regeneration using the MOCART 2.0 knee score

2.5. Arthroscopic evaluations

The MFC articular cartilage was evaluated arthroscopically before and approximately 1 year after HTO and hUCB-MSCs implantation. At the time of the initial HTO, cartilage lesions in the medial compartment were assessed and graded using the ICRS system, which classifies cartilage conditions as follows: grade 0, normal cartilage; grade 1, nearly normal cartilage with superficial fissures or softening but no considerable loss of thickness; grade 2, defects involving <50% of the cartilage thickness; grade 3, lesions extending beyond 50% of the thickness, potentially reaching the calcified layer (3B) or the subchondral bone (3C) without penetration into the subchondral bone; and grade 4, full-thickness cartilage loss with subchondral bone involvement [34]. The size and severity of the MFC defects were also documented.

At approximately 1 year postoperatively, plate removal and a second-look arthroscopy were routinely performed to evaluate cartilage regeneration. The decision to perform these interventions was further supported by substantial coverage provided by the National Health Insurance Service, facilitating their implementation [35,36]. The quality of cartilage regeneration was assessed using the ICRS Cartilage Repair Assessment (CRA), which classifies outcomes as follows: grade I, normal; grade II, nearly normal; grade III, abnormal; and grade IV, severely abnormal. The ICRS CRA evaluates the degree of defect fill, level of tissue integration with the surrounding native cartilage, and macroscopic appearance of the repaired surface. Scores range from 0 to 12, with higher scores indicating superior cartilage repair quality [9,37,38]. Rigidity of the repaired cartilage upon probing was also assessed and classified as rigid, soft, and very soft compared with adjacent cartilage, in line with the Oswestry Arthroscopy Score system [11,39]. This comprehensive evaluation provides detailed insights into the cartilage repair quality following the surgical intervention.

2.6. Statistical analysis

Parametric variables, e.g., clinical scores, MOCART score, and graft extrusion, were compared between the groups using the independent t-test, whereas non-parametric variables, including the K–L grade and ICRS grade, were analyzed using the Mann–Whitney U test. Preoperative and postoperative comparisons for parametric variables were conducted using the paired t-test, and non-parametric variables were compared using the Wilcoxon signed-rank test. Categorical variables were analyzed using either the Pearson chi-square or Fisher exact test, as appropriate. Spearman rank correlation analysis was performed to evaluate the relationship between preoperative MM extrusion and the state of cartilage regeneration. Intra-observer and inter-observer reliabilities were assessed using intraclass correlation coefficients (ICCs),

categorized as excellent (0.76–1.00), fair to good (0.40–0.75), or poor (0–0.40). Statistical analyses were performed using SPSS software (IBM SPSS Statistics 21; IBM Corp, Armonk, NY, USA), with statistical significance defined as $p < 0.05$.

The sample size calculation was performed using a two-sided superiority test, taking into account the IKDC subjective score. The analysis assumed a significance level (α) of 5%, a statistical power of 90%, and a dropout rate of 10%. The minimal clinically important difference (MCID) for the IKDC score was estimated to be 16.7, with a standard deviation (SD) of 13, based on previously published data. As a result, at least 15 participants per group were deemed necessary to achieve adequate statistical validity [11,40,41].

3. Results

Table 1 presents the patients' demographic and baseline characteristics. Both groups had similar preoperative clinical characteristics, excluding MM extrusion.

At the final follow-up, the IKDC subjective and WOMAC scores showed significant improvement in both groups (all, $p < 0.001$). There were no statistically significant differences in the postoperative IKDC subjective scores between groups P (77.3 ± 10.7) and L (73.1 ± 11.8) ($p = 0.207$). Similarly, the postoperative WOMAC scores did not differ significantly between groups P (15.4 ± 7.3) and L (17.2 ± 7.7) ($p = 0.415$). The isokinetic strength test result was not significantly different between the groups. The isokinetic extensor strengths at 60° were 121.5 ± 24.3 Nm/kg in group P and 118.2 ± 28.4 Nm/kg in group L ($p = 0.669$). Additionally, the limb symmetry index for extensor strength was similar between groups P (0.8 ± 0.2) and L (0.8 ± 0.3) ($p = 0.999$), indicating no statistically significant differences in muscle strength between the groups (Table 2).

At the 1-year follow-up MRI assessment, there was a significant difference in MM extrusion between the groups, with group L showing greater extrusion (5.3 ± 1.3 mm) than group P (4.2 ± 1.1 mm) ($p = 0.016$). Additionally, the incidence of pathological MM extrusion was significantly higher in group L (100%) than in group P (76%) ($p = 0.023$), indicating a greater tendency for pathologic extrusion in group L.

Table 1
Demographic and baseline characteristics of patients in groups P and L.

	Group P (n = 25)	Group L (n = 22)	p-value
Age, years	56.8 ± 5.4	57.6 ± 5.9	0.553
BMI, kg/m ²	25.1 ± 2.2	26.2 ± 2.9	0.147
Sex, n (male/female)	6/19	7/15	0.550
Clinical follow-up period, months	30.5 ± 4.7	31.3 ± 3.9	0.532
MRI follow-up duration, months	13.1 ± 1.8	12.7 ± 1.6	0.428
Second-look and plate removal, months	13.1 ± 1.6	12.6 ± 1.7	0.305
Preoperative IKDC subjective score	52.3 ± 11.3	48.7 ± 12.4	0.303
Preoperative WOMAC score	45.6 ± 6.4	47.2 ± 6.1	0.387
Preoperative mechanical axis (hip-knee-ankle angle), °	Varus	Varus	0.154
Preoperative Kellgren–Lawrence grade (I/II/III/IV), n	6.6 ± 1.1	7.1 ± 1.2	0.302
Preoperative ICRS grade of MFC, n (%)	0/14/11/0	0/9/13/0	0.901
Grade 3C	5 (20%)	4 (18.2%)	
Grade 4A	18 (72%)	15 (70.2%)	
Grade 4B	2 (8%)	3 (10.6%)	
MFC lesion size, cm ²	6.2 ± 2.3	6.9 ± 2.1	0.2854
Preoperative MM extrusion, mm	3.4 ± 0.6	4.3 ± 1.2	0.002
Preoperative pathological MM extrusion, n (%)	16 (64%)	19 (86.4%)	0.079

BMI, body mass index; MRI, magnetic resonance imaging; IKDC, International Knee Documentation Committee; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; ICRS, International Cartilage Repair Society; MFC, medial femoral condyle; MM, medial meniscus.

Table 2

Clinical and radiological outcomes of patients in groups P and L.

	Group P (n = 25)	Group L (n = 22)	p- value
IKDC subjective score	77.3 ± 10.7	73.1 ± 11.8	0.207
WOMAC score	15.4 ± 7.3	17.2 ± 7.7	0.415
Isokinetic extensor strength at 60 ° (Nm/kg)	121.5 ± 24.3	118.2 ± 28.4	0.669
LSI for extensor strength (%)	0.8 ± 0.2	0.8 ± 0.3	0.999
Mechanical axis (hip-knee-ankle angle), °	Valgus 1.8 ± 1.0	Valgus 2.1 ± 1.1	0.393
Postoperative MM extrusion, mm	4.2 ± 1.1	5.3 ± 1.3	0.016
Pathologic MM extrusion, n (%)	19 (76%)	22 (100%)	0.023

IKDC, International Knee Documentation Committee; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; LSI, limb symmetry index; MM, medial meniscus.

Group P showed higher MOCART 2.0 scores than group L; however, this difference did not reach statistical significance (56.8 ± 10.5 versus [vs.] 54.1 ± 8.8 , $p = 0.347$). Although there were no significant differences between the groups across the individual parameters of the MOCART 2.0 score, a higher proportion of patients in group P showed no major subchondral changes or only minor edema-like marrow signal compared to group L (56% vs. 31.8%, $p = 0.096$) (Table 3).

There were no significant differences in the total ICRS CRA score between groups P and L during the second-look evaluation (9.4 ± 1.8 and 8.6 ± 1.7 , respectively; $p = 0.083$) (Fig. 3). Similarly, no significant differences were observed in any individual parameters of the ICRS CRA. Although group P had a greater proportion of patients with ICRS CRA grades I or II than group L (84% and 72.7% respectively; $p = 0.480$), this difference was not statistically significant. Furthermore, the assessment of the rigidity of the regenerated cartilage tissue, as evaluated through probing, showed no significant differences between the groups, with 68% and 54.5% of patients in groups P and Group L, respectively, demonstrating rigid cartilage tissue ($p = 0.565$) (Table 3). The ICCs for intra-observer reliability were 0.88 and 0.89 for the MOCART and ICRS CRA scores, respectively, and those for inter-observer reliability were 0.85 and 0.87, respectively, demonstrating excellent reliability.

The correlation analysis revealed no significant association between preoperative MM extrusion and the ICRS CRA score ($r = -0.211$, $p = 0.053$). However, it showed a significant negative correlation between preoperative MM extrusion and both the MOCART 2.0 score ($r = -0.237$, $p = 0.029$) and subchondral changes, a parameter within the MOCART score ($r = -0.285$, $p = 0.017$). These findings suggest that greater preoperative meniscal extrusion is associated with increased subchondral changes following cartilage regeneration.

Table 3

Cartilage regeneration of the medial femoral condyle on mri and second-look arthroscopy.

	Group P (n = 25)	Group L (n = 22)	p- value
MOCART 2.0 score	56.8 ± 10.5	54.1 ± 8.8	0.347
ICRS CRA score	9.4 ± 1.8	8.6 ± 1.7	0.083
ICRS CRA grade, n (%)			0.626
Grade I	4 (16%)	2 (9.1%)	
Grade II	17 (68%)	14 (63.6%)	
Grade III	4 (16%)	6 (27.3%)	
Grade IV	0	0	
ICRS CRA grade I or II, n (%)	21 (84%)	16 (72.7%)	0.480
Rigidity of repaired tissue, n (%)			0.565
Rigid	17 (68%)	12 (54.5%)	
Softer	7 (28%)	8 (36.4%)	
Very soft	1 (4%)	2 (9.1%)	

MRI, magnetic resonance imaging; MOCART, Magnetic Resonance Observation of Cartilage Repair Tissue; ICRS CRA, International Cartilage Repair Society Cartilage Repair Assessment.

4. Discussion

The principal finding of the present study was the significant improvement in clinical outcomes following HTO combined with hUCB-MSC implantation, as demonstrated by notable increases in IKDC and WOMAC scores. Importantly, cartilage regeneration was satisfactory regardless of the preoperative condition of the MM, as confirmed by MRI and second-look arthroscopy, which contrasts with our study hypothesis. However, the preoperative state of the MM, particularly in cases where the hoop tension of the meniscus was compromised, may influence postoperative subchondral bone integrity, potentially affecting long-term clinical and structural outcomes.

Addressing large cartilage defects in OA requires a comprehensive approach due to the combined involvement of cartilage and subchondral bone pathology. hUCB-MSCs have demonstrated promising chondrogenic and immunomodulatory properties, providing a dual approach to cartilage repair and subchondral bone remodeling [42–45]. Studies, e. g., those by Jin et al. [46], have demonstrated the efficacy of bone marrow-derived MSCs and their exosomes in mitigating cartilage destruction and improving subchondral bone architecture, restoring key structural features, such as trabecular bone volume and connectivity. In this context, allogenic hUCB-MSCs have emerged as a promising therapeutic option because of their potent chondrogenic properties and immunomodulatory effects. These cells not only facilitate cartilage regeneration but also play a crucial role in subchondral bone remodeling, addressing both aspects of OA pathology [47,48]. Comparative studies have demonstrated the superiority of hUCB-MSCs over traditional methods such as microfracture or microdrilling when combined with HTO [8–11,49,50]. Histologic assessments and long-term clinical outcomes at the 3–5-year follow-up were significantly better in the hUCB-MSC group than in the microfracture group ($p < 0.05$), with no significant differences in adverse events between the groups [49]. These results underscore the potential of hUCB-MSCs to treat cartilage and subchondral bone degeneration, offering a comprehensive solution for OA-related cartilage defects.

Here, the mean total MOCART 2.0 score was 55.5 ± 9.7 , with a mean ICRS CRA score of 9.0 ± 1.8 . In group P, where meniscal integrity was better preserved, the mean total MOCART 2.0 score was 56.8 ± 10.5 , and the average ICRS CRA score was 9.4 ± 1.8 . A recent prospective study comparing hUCB-MSC implantation with microdrilling during HTO reported a mean total MOCART 2.0 score of 50.3 ± 12.6 and an ICRS CRA score of 8.5 ± 2.1 in the hUCB-MSC implantation group, which are lower than the mean values observed in group L in our study. Although the patient cohorts are not large enough for direct comparison, our study used a 10-mm drilling depth for hUCB-MSC implantation compared with the conventional depths of 4–7 mm. This deeper drilling has been hypothesized to improve stem cell integration by providing greater access to the bone marrow and allowing for the implantation of a higher number of stem cells within the hole itself, thereby enhancing the effectiveness of MSCs in cartilage repair [10,15,48,49]. Thus, our approach might enhance cartilage regeneration quality and addresses the underlying subchondral bone pathology in OA [51]. Chen et al. [52] demonstrated that deeper drilling (6 mm vs. 2 mm or microfracture) considerably enhances cartilage repair by improving access to bone marrow, which is essential for promoting regenerative processes. Kim et al. [53] reported that the quantity of implanted MSCs is a key predictor of clinical and MRI outcomes, with higher MSC numbers associated with improved cartilage regeneration.

Despite the growing research on HTO combined with hUCB-MSC implantation, few studies have examined outcomes based on the condition of the MM at the time of surgery [11,49,54]. Here, there were no significant differences in total MOCART 2.0 or ICRS CRA scores between the groups with and without preserved meniscal integrity at the time of surgery. However, correlation analysis revealed that preoperative MM extrusion was significantly negatively correlated with the total MOCART 2.0 score and the subchondral change score. This trend suggests a more

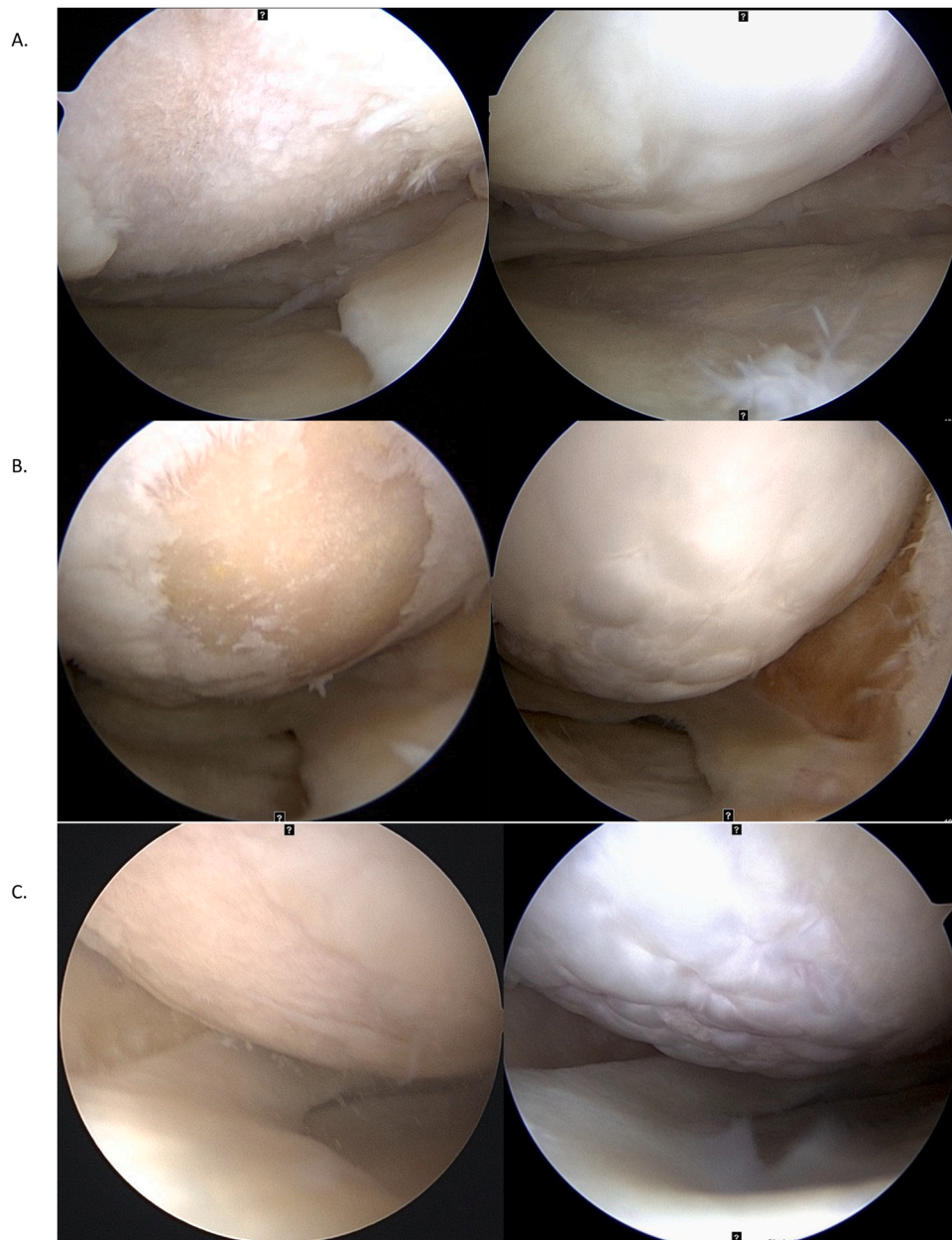


Fig. 3. Evaluation of repaired cartilage according to International Cartilage Repair Society (ICRS) cartilage repair assessment (CRA). **A:** A case of ICRS CRA score of 12, which was grade I (normal) in group L. **B:** A case of ICRS CRA score of 10, which was grade II (nearly normal) in group P. **C:** A case of ICRS CRA score of 6, which was grade III (abnormal) in group P.

favorable subchondral response in group P, although the difference was not statistically significant. This may potentially affect long-term outcomes, as subchondral health is crucial for sustained cartilage repair and joint function. Understanding the influence of the MM status is crucial, as it plays an important role in joint biomechanics and cartilage biology. Choe et al. [17] reported varying degrees of meniscal integrity in patients undergoing HTO: nine had intact menisci, 20 underwent partial meniscectomies, and 30 had subtotal meniscectomies. Despite significant clinical improvements observed postoperatively in all groups, patients with subtotal meniscectomies experienced a marked reduction in joint space width at midterm follow-up (≥ 4 years) compared with those without meniscal tears. These findings emphasize the crucial importance of preserving as much of the meniscus as possible during HTO to maintain optimal joint biomechanics. Consequently, some studies have been conducted comparing groups undergoing HTO with concurrent medial meniscal root repair to those receiving only HTO. This approach

is increasingly adopted to enhance joint stability and improve long-term outcomes by addressing alignment and meniscal integrity simultaneously, which are crucial for optimal cartilage preservation and overall knee function. Additionally, Park et al. [21] demonstrated that medial meniscal root repair consistently reduces contact pressure and increases the contact area within the joint, regardless of the angular correction achieved through HTO. This biomechanical benefit persists across various correction angles during HTO, emphasizing the importance of meniscal root repair in maintaining joint stability and functionality. Moreover, Choi et al. [22] found that medial meniscal posterior root repair combined with HTO enhanced root tear healing and cartilage regeneration more effectively than HTO alone, and this improvement was observed regardless of the degree of correction or the application of microfracture.

This study has several limitations. First, as a retrospective comparative study, it is inherently subject to selection bias, which may affect

the comparability of the two groups. Second, despite calculating the sample size based on an appropriate power analysis, the small number of participants in each group may restrict the generalizability of the findings and reduce the statistical power to detect subtle differences in outcomes. Third, the follow-up period, although adequate for short-term clinical and radiological assessments, may not have been sufficient for evaluating long-term outcomes, particularly regarding cartilage regeneration durability and subchondral bone integrity. Lastly, while statistical methods, including parametric and non-parametric tests, were appropriately employed, the potential for residual confounding remains. Thus, future studies with a prospective design and larger sample size are needed to confirm these results.

5. Conclusion

HTO with hUCB-MSC implantation provided significant clinical improvements and effective cartilage regeneration regardless of preoperative MM status. However, preoperative MM extrusion may influence subchondral bone changes, emphasizing the need to consider MM status for long-term outcomes.

CRediT authorship contribution statement

Conceptualization, D.-W.L.; methodology, S.-G.M. and S.-W.H.; formal analysis, J.-H.K., S.-G.M., and S.-W.H.; investigation, S.-W.H., J.-H.K., and S.-I.C.; data curation, D.-W.L. and S.-G.M.; writing—original draft preparation, D.-W.W. and S.-W.H.; writing—review and editing, D.-W.L.; supervision, D.-W.L. and S.-G.M. All authors have read and agreed to the published version of the manuscript.

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